

PROJECT REPORT ON
Neuro-Eye controlled Toy car

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE AWARD
OF

**DIPLOMA
IN
COMPUTER ENGINEERING**

Submitted by:-

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DEPARTMENT OF COMPUTER ENGINEERING

GOVERNMENT POLYTECHNIC, NAGPUR
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2024-25

CERTIFICATE

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This is certified that,

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Of term Odd-2024 of Computer Engineering department students have submitted their Project report on “**Neuro-Eye controlled Toy car**” during academic session 2024-2025 as a part of project work prescribed by Government Polytechnic, Nagpur for the partial fulfillment of the requirement of diploma in Computer Engineering, Fifth Semester. The project work is the record of student own work and is completely satisfactory.

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**INTERNAL
EXAMINER**

**EXTERNAL
EXAMINER**

DATE:

GOVERNMENT POLYTECHNIC, NAGPUR

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DEPARTMENT OF COMPUTER ENGINEERING

CANDIDATE'S DECLARATION

We, students of Diploma (Computer Engineering Department), hereby declare that the project Report titled “**Neuro-Eye controlled Toy car**” which is submitted by me to the Department of Computer Science & Engineering, **Government Polytechnic, Nagpur** in partial fulfillment of the requirement for the award of the Diploma is original and not copied from any source without proper citation. This work has not previously formed the basis for the award of and Degree, Diploma Associate ship, Fellowship or other similar title or recognition.

Place: Nagpur

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The success of any work depends on efforts of many individuals. We would like to take this opportunity to express our deep gratitude to those who extended their support and have guided us to complete this project work.

We wish to express our sincere and deepest gratitude to our guide **Mrs. D.S Nalinde** for her invaluable and unique guidance. We would also like to thank her for the constant source of help, inspiration and encouragement in the successful completion of project. It has been our privilege and pleasure to work under her expert guidance.

We like to thank **Dr. M. V Sarode**, Head of Department, for providing us the necessary information about the topic. We would again like to thank **Dr. R.P Mogre**, Principal of our College, for providing us the necessary help and facilities we needed.

We express our thanks to all the staff members of Computer Engineering department who have directly or indirectly extended their kind co-operation in the completion of our project Report.

ABSTRACT

The Neuro-Eye Controlled Toy Car initiative leverages Brain-Computer Interface (BCI) and Electrooculography (EOG) technologies to create an innovative dual-input assistive control system. This system is designed to help individuals with mobility impairments by enabling control through brainwaves and eye movements. EEG signals are used to execute forward and stop commands, while EOG signals detect left and right eye movements (horizontal) to navigate directionally. The integration of these non-invasive technologies provides an intuitive and accessible interface for users with limited physical capabilities.

The system employs the BioAmp EXG Pill and an Arduino microcontroller to acquire, preprocess, and classify EEG and EOG signals in real-time. Machine learning algorithms are utilized for EEG signal analysis, while threshold-based detection is applied for EOG signals. The processed commands are transmitted to control the movement of a toy car, demonstrating the feasibility of this approach. The simplicity of its design ensures ease of use, making it suitable for individuals with diverse physical challenges.

While currently implemented on a toy car, the system's scalability offers potential for broader applications, such as electric wheelchairs and robotic assistance. This adaptability highlights its capability to revolutionize assistive technology by providing greater independence and enhanced functionality for users.

The Neuro-Eye Controlled Toy Car showcases the potential of combining BCI and EOG technologies to create intelligent, user-centric assistive devices. By bridging the gap between neuroscience and mobility solutions, this initiative takes a significant step toward improving the quality of life for individuals with disabilities, paving the way for future advancements in assistive technologies.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

As technology advances, integrating Brain-Computer Interface (BCI) systems with assistive devices is becoming increasingly viable. BCIs create a direct connection between the brain and external devices, offering groundbreaking solutions for individuals with physical disabilities. The "Neuro-Eye Controlled Toy Car" initiative leverages BCI technology to allow users to control a toy car using their brain activity and eye movements, demonstrating how non-invasive methods can translate neural signals into real-world actions.

In this system, electroencephalography (EEG) signals are captured using the BioAmp EXG Pill and processed by an Arduino microcontroller to interpret specific brainwave patterns. Through EEG, users can issue commands to move the car forward or stop, while electrooculography (EOG) tracks eye movements to control the car's left and right turns. The integration of these two technologies offers an intuitive and user-friendly way for individuals to interact with the device, highlighting the potential of combining brainwave and eye movement tracking for assistive control.

This toy car system serves as a proof of concept for broader applications, such as in electric wheelchairs and other assistive devices. By demonstrating the feasibility of controlling movement through thought and eye control, this initiative aims to inspire future developments in BCI and EOG technologies, ultimately empowering individuals with mobility impairments and improving their independence.

1.2 Problem Statement

Individuals with mobility impairments often face challenges using traditional assistive devices that require physical interaction, such as joystick controls or switches. These limitations restrict their independence and make it difficult to fully interact with their environment.

Combining Brain-Computer Interface (BCI) technology with Electrooculography (EOG) provides a potential solution by enabling users to control devices using brain activity and eye movements, bypassing the need for traditional physical controls. This system can address the shortcomings of current devices and significantly enhance mobility and independence for people with disabilities.

1.3 Solution

This project proposes the development of a dual-input BCI system that enables users to control a toy car using both electroencephalography (EEG) and electrooculography (EOG) signals. The system utilizes the BioAmp EXG Pill to capture EEG and EOG signals, and an Arduino microcontroller to process the signals and execute commands. The EEG signals are used to control forward and stop movements, while the EOG signals allow the user to control left and right turns by tracking eye movements.

This system aims to provide an easy-to-use interface for users with mobility impairments, enabling them to control the toy car by simply thinking or moving their eyes. This dual-input approach demonstrates the potential of combining BCI and EOG technologies for assistive control, offering a glimpse into the future of more sophisticated mobility aids, such as electric wheelchairs and other assistive devices, thus improving the quality of life for individuals with physical disabilities.

1.4 Project Objective

The primary objective of this research is to develop and validate a Neuro-Eye Controlled Toy Car system that integrates both EEG and EOG signals for controlling the car. The specific objectives of this initiative are:

- **Develop the NeuroEye System:** Create a dual-input BCI system that integrates EEG signal acquisition and processing, using the BioAmp EXG Pill and Arduino microcontroller to control a toy car. The system will interpret EEG signals for forward and stop commands and EOG signals for left and right turns.
- **Validate Real-Time Processing:** Ensure the system's real-time performance in translating EEG and EOG signals into actionable commands with minimal latency, ensuring reliable control over the car's movements.
- **Support Assistive Technologies:** Provide a functional prototype of a Neuro-Eye controlled device that can be adapted for more complex assistive technologies like electric wheelchairs, robotic arms, or other mobility aids. This will demonstrate the potential of BCI and EOG technologies in empowering individuals with physical disabilities to gain greater independence.

1.5 Scope of Study

This study aims to develop a Neuro-Eye Controlled Toy Car system that allows users to control the toy car using EEG and EOG signals. The research will focus on integrating the BioAmp EXG Pill for EEG and EOG signal acquisition, with an Arduino microcontroller for signal processing. The study will involve:

- **Signal Optimization:** Exploring various electrode configurations to optimize the detection of brainwave patterns and eye movements.
- **User Testing:** Assessing the usability and accessibility of the system for individuals with mobility impairments, ensuring the interface is intuitive and user-friendly.
- **Future Applications:** Investigating the feasibility of scaling this system for broader assistive technologies, such as electric wheelchairs and other mobility aids.

The study will also assess the effectiveness of combining EEG and EOG signals in providing reliable control over the toy car, offering insights into the practical applications of these technologies in assistive devices.

1.6 Project Significance

The significance of this study lies in its potential to enhance the development of Brain-Computer Interface (BCI) technology as a practical solution for individuals with mobility impairments. By demonstrating the feasibility of controlling a toy car using EEG and EOG signals, the research provides a proof of concept for future non-invasive BCI systems that can control devices without the need for physical interaction.

This project highlights the practicality of combining BCI and EOG technologies for assistive control, paving the way for more sophisticated mobility aids, such as electric wheelchairs, that could greatly improve the quality of life for individuals with disabilities. By contributing to the growing body of knowledge in BCI applications, this study encourages further innovation and research in the field, ultimately empowering individuals with disabilities and enhancing their independence.

CHAPTER 2

LITERATURE SURVEY

2.1. Brain-Computer Interface (BCI) for Assistive Devices

Wolpaw et al. (2002), as referenced in “Recent Advances in Brain-Computer Interfaces” by Ulrich Hoffmann et al.[1], established the feasibility of using electroencephalography (EEG) for non-invasive brain signal acquisition. Their system primarily utilized EEG caps with electrodes placed on the scalp to record brainwave patterns, translating them into control signals for external devices like cursors, robotic arms, or mobility aids. This study employed basic signal processing techniques like Common Spatial Patterns (CSP) and Linear Discriminant Analysis (LDA) to classify brain signals and distinguish between mental states (e.g., imagining moving the left or right hand). Despite its pioneering nature, the study faced challenges in signal quality due to electrode placement issues, noise from muscle activity, and external interference, reducing the reliability of the BCI systems.

2.2. Real-Time EEG and EOG Signal Processing in Hybrid Systems

Kumar et al. (2018), discussed in “Hybrid Brain-Computer Interface (BCI) based on the EEG and EOG signals” by Jun Jiang et al.[2], developed a dual-input control system integrating EEG for brainwave detection and EOG for eye movement tracking. This system, aimed at controlling prosthetic devices, utilized EEG headsets (e.g., Emotiv EPOC) and EOG electrodes placed near the eyes, enhancing versatility by combining mind-based (EEG) and eye gesture-based (EOG) control. However, the system faced challenges like signal synchronization issues, increased computational load, and user fatigue when managing dual inputs. Lack of adaptive filtering techniques to eliminate noise further impacted signal consistency, particularly in varied conditions.

2.3. EEG-Based Control Systems for Mobility Applications

Millán et al. (2004), referenced in “Progress in Brain Computer Interface: Challenges and Opportunities” by Simanto Saha et al.[3], developed an EEG-based wheelchair control system using motor imagery. EEG caps with wet electrodes detected brain activity related to movement intentions, with control based on recognizing specific brainwave patterns (such as mu and beta rhythms). Signal processing algorithms like Independent Component Analysis (ICA) were used to isolate relevant signals. Despite its promise, the system struggled with real-time processing, signal fidelity, and robustness in real-world scenarios. Challenges like latency, subtle brain signal detection, bulky hardware, and complex user-specific calibration further limited its practicality.

2.4. EOG-Based Control for Directional Navigation

Barea et al. (2002), as cited in “EOG-Based Human–Computer Interface: 2000–2020 Review” by Chama Belkhiria et al.[4], explored EOG signals for controlling assistive devices, focusing on directional navigation. Using electrodes placed near the eyes, the system detected subtle eye movements and translated them into commands for wheelchair control, enabling left and right turns through horizontal and vertical eye movement detection. Challenges included eye fatigue during prolonged usage, sensitivity to lighting conditions, and head position variability. The lack of adaptive algorithms to adjust for individual differences in eye movement patterns also affected accuracy and reliability in dynamic environments.

2.5. Advancements in BCI and EOG Integration for Enhanced Control

Pfurtscheller et al. (2010), in alignment with “Brain–computer interface: trend, challenges, and threats” by Baraka Maiseli et al.[5], developed a hybrid BCI-EOG system combining EEG and EOG data for improved control accuracy. EEG caps detected brainwave signals, while EOG electrodes tracked eye movements, resulting in enhanced precision and control for assistive devices. Despite improvements, challenges like signal interference, increased computational load, and system complexity persisted. Issues related to signal drift and environmental sensitivity occasionally led to inaccuracies, while the hybrid setup's cost and portability limited its broader adoption.

2.6. Multi-Way Controlled Robot Vehicle Using Arduino and RF Module

In “Multi-Way Controlled Robot Vehicle Using Arduino and RF Module” by Syed Mazhar Ali Shah[6], the author presents a project focused on developing a robot vehicle controlled via RF modules using Arduino. The system allows for multi-way control of the robot, enabling movement in different directions, making it more versatile compared to traditional remote-controlled vehicles. The design is based on low-cost components, making it an accessible solution for educational and practical purposes. However, the system relies on RF communication, which can face issues like interference and range limitations. Despite these challenges, the project provides valuable insights into developing efficient and affordable robotic systems using basic electronic components and RF technology. This is relevant to our project, where we are also working on controlling a robotic vehicle (toy car) but with a brain-computer interface (BCI) for control, offering a more intuitive and innovative method of operation.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 Analysis

Analysis is a process of collecting and interpreting facts, identifying the problems, and decomposition of a system into its components. System analysis is conducted for the purpose of studying a system or its parts to identify its objectives.

3.1.1 Hardware Requirements

No.	HARDWARE COMPONENTS
1	Arduino Uno Microcontroller
2	BioAmp EXG Pill (2)
3	BioAmp cables
4	Electrodes
5	Jumper cables
6	Conductive gel
7	Computer System
8	RF Module
9	Motor Driver
10	HF12D (Decoder), HT12E(Encoder)
11	Toy car

1. Arduino Uno Microcontroller



Figure 3.1.1: Arduino Uno Microcontroller

The Arduino Uno is the core processing unit, responsible for controlling the toy car based on brainwave and eye movement signals received from the BioAmp EXG Pills. It processes EEG signals for forward and backward movement and EOG signals for left and right movement.

2. BioAmp EXG Pill

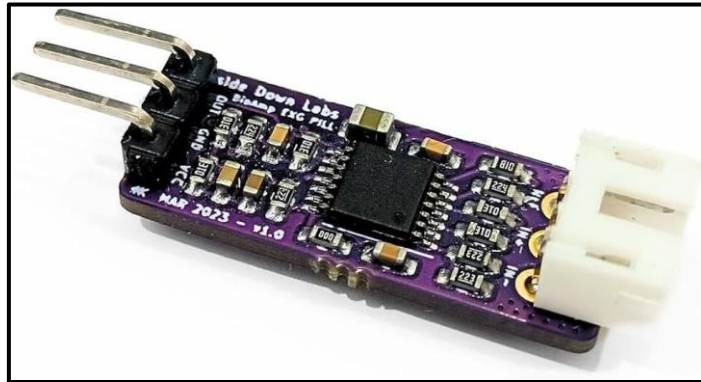


Figure 3.1.2 3.1.2 BioAmp EXG Pill

The BioAmp EXG Pill acquires EEG signals, specifically capturing brainwave data related to forward and stop movements. It is connected to the scalp with electrodes for real-time EEG signal collection.

The second BioAmp EXG Pill is used for acquiring EOG signals from the eyes, which are critical for detecting left and right movement commands. Electrodes placed near the eyes capture electrical signals corresponding to eye movements, which are then interpreted by the system.

3. BioAmp cables

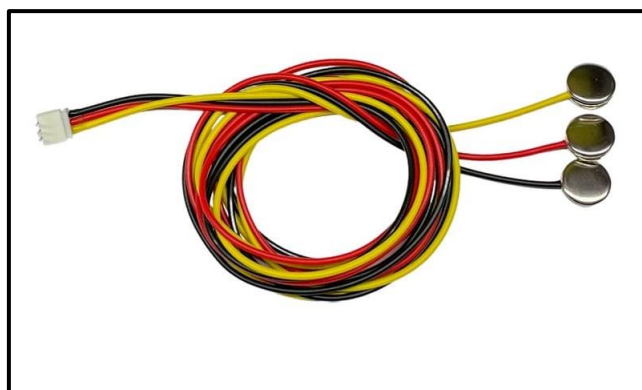


Figure 3.1.3 BioAmp cables

The BioAmp cables connect the BioAmp EXG Pills to the electrodes, ensuring accurate and stable signal transmission for both EEG and EOG data.

4. Electrodes



Figure 3.1.4 Electrodes

Electrodes are used for detecting the electrical activity generated by both brainwaves and eye movements. Wet electrodes with conductive gel are required for proper signal quality.

5. Jumper cables

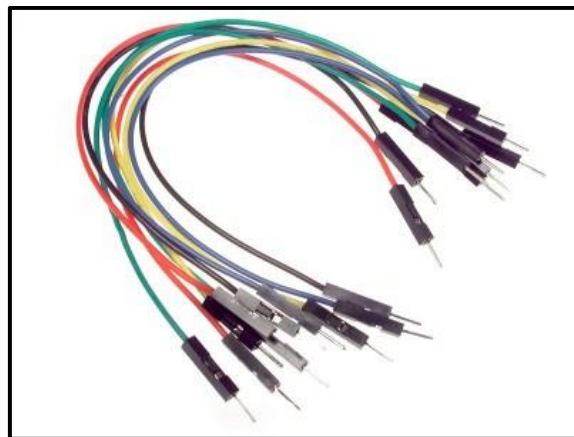


Figure 3.1.5 Jumper cables

Jumper cables are flexible wires with connectors on both ends, commonly used for establishing connections in electronic projects. They facilitate the connection between various components, such as the BioAmp EXG Pill and Arduino, ensuring reliable data transfer and control signals within the system.

6. Conductive Gel



Figure 3.1.6 Conductive Gel

Conductive gel is applied to the electrodes to improve the contact between the electrodes and skin, ensuring that the EEG and EOG signals are captured with minimal noise.

7. Computer System

A computer system is necessary for developing and testing. It can also be used for data analysis, visualization, and monitoring the performance of the system in real-time, ensuring the efficient functioning of the entire setup.

8. RF module

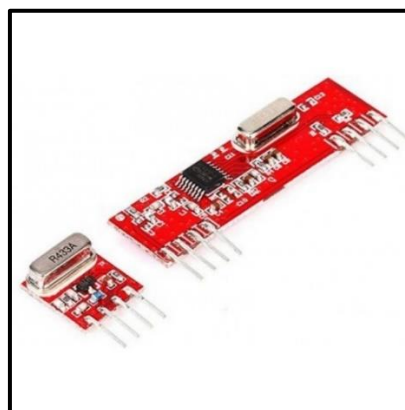


Figure 3.1.8 RF module

Facilitates wireless communication between the controller (Arduino) and the toy car. It transmits the encoded control signals to ensure seamless command execution over a distance.

9. Motor driver

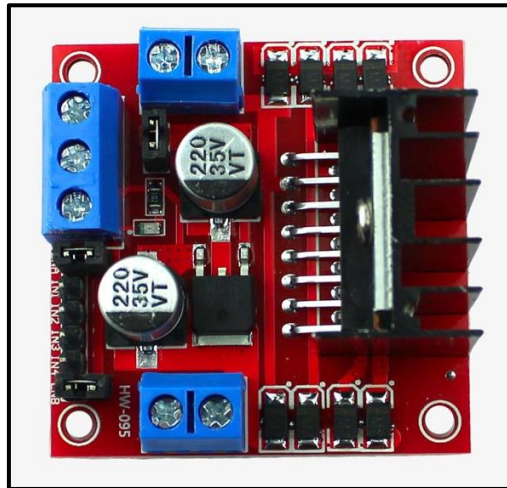


Figure 3.1.9 Motor driver

Acts as an interface between the microcontroller and the motors of the toy car. It receives control signals and provides the necessary voltage and current to drive the motors in the desired direction.

10. HF12D (Decoder) and HT12E (Encoder)

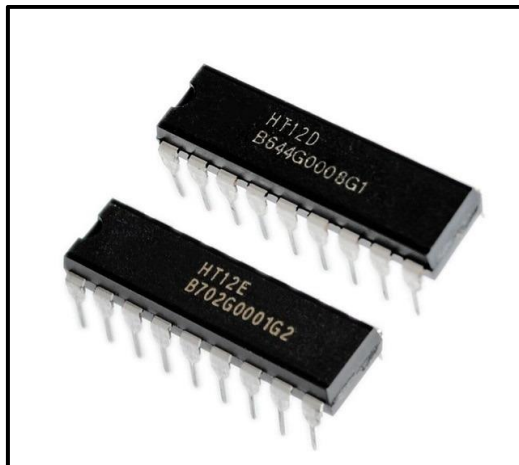


Fig 3.1.10 HF12D (Decoder) and HF12E (Encoder)

HF12D (Decoder): Decodes the signals received from the RF Module into a readable format for the motor driver. It ensures that the car performs actions based on the correct commands (forward, stop, left, or right).

HF12E (Encoder): Encodes the commands from the Arduino into a format suitable for transmission by the RF Module. This encoding ensures accurate data transfer without loss or corruption.

11. Toy Car

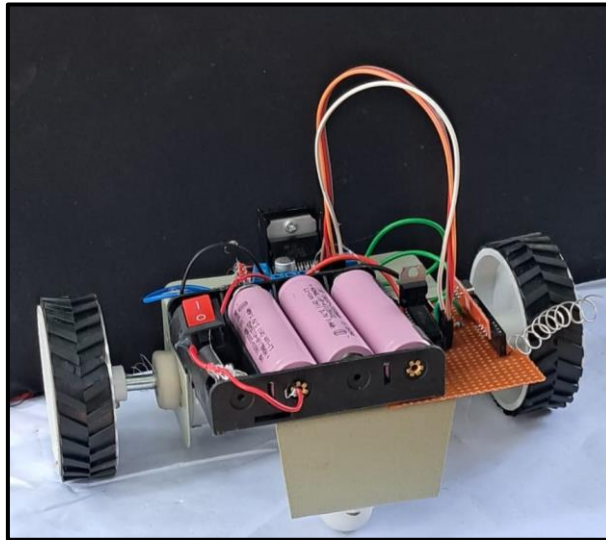


Figure 3.1.11 Toy Car

The end device that executes the received commands. It moves forward, stops, or turns left/right based on the signals processed and decoded by the system.

3.1.2 Software Requirements:

Here are the required software :-

No.	SOFTWARE COMPONENTS
1	Arduino IDE
2	Visual Studio Code
3	Star UML
4	Python

Table 3.2.1 Software Components

1. Arduino IDE

- Purpose: Used for programming the Arduino Uno microcontroller.
- Functionality: Enables the development of efficient and tailored code for the microcontroller, ensuring seamless integration with the hardware components.

2. Visual Studio Code

- Purpose: A versatile code editor for broader system development tasks.
- Functionality: Offers a user-friendly environment for writing, editing, and debugging code. Its extensibility allows integration with various programming languages and tools.

3. Star UML

- Purpose: Tool for preparing the system design.
- Functionality: Star UML is a visual modelling tool that allows system architects and designers to create detailed diagrams and models of the system's architecture. It aids in visualizing and communicating the structure and relationships between various components.

4. Python

- Purpose: Python is employed in this project for data processing and algorithm development, enabling the effective interpretation of signals. Its simplicity and versatility facilitate real-time processing for controlling the toy car.
- Functionality: Python's robust libraries support various project components, allowing for efficient signal processing and seamless integration of machine learning models to translate data into commands.
- Required Libraries:
 - NumPy
 - Pandas
 - sklearn (Scikit-learn)
 - Serial
 - SciPy
 - Time
 - Pickle

3.1.3 Functional Requirements

The NeuroEye -controlled toy car must satisfy the following functional requirements:

1. **EEG Signal Acquisition:** The system captures real-time EEG signals using the BioAmp EXG Pill (1st), providing control for forward and stop movements.
2. **EOG Signal Acquisition:** The second BioAmp EXG Pill captures EOG signals,

translating eye movements into left and right commands for controlling the toy car.

3. **Signal Filtering and Processing:** The system filters both EEG and EOG signals to remove noise and extract meaningful patterns for command interpretation.
4. **Command Translation:** EEG patterns translate into forward and stop commands, while EOG patterns translate into left and right commands.
5. **User Interaction:** The interface allows the user to initiate signal acquisition and visualize the control commands in real time.
6. **Feedback Mechanism:** The system provides feedback about the toy car's movement, indicating whether it is moving forward, backward, left, right, or stopped.
7. **Low Latency Operation:** The system ensures minimal latency in processing the signals, allowing smooth operation of the toy car.

3.1.4 Non- Functional Requirements

Non-functional requirements specify the overall qualities and constraints that the system must meet, ensuring it operates effectively in a real-world environment. The BCI-controlled toy car system must meet the following non-functional requirements:

1. **Performance:** The system must process EEG and EOG signals with minimal latency (under 200 ms) to ensure responsive control.
2. **Reliability:** The system should consistently capture and interpret EEG and EOG signals with high accuracy, minimizing errors in command execution.
3. **Usability:** The user interface should be simple and intuitive for individuals of varying technical expertise.
4. **Scalability:** The system should be capable of supporting future enhancements, such as controlling other devices (e.g., robotic arms, wheelchairs).
5. **Safety:** The system should incorporate fail-safes to prevent the toy car from operating under unsafe conditions, ensuring user safety.
6. **Compatibility:** The system should be compatible with various computer systems and software tools for monitoring and data analysis.
7. **Maintainability:** The system should be easy to maintain and update, with provisions for troubleshooting and adding new features based on user feedback.

3.1.5. Feasibility Study

A feasibility study is an analysis that assesses the practicality and viability of a proposed project or system. It evaluates various aspects to determine whether the project should proceed. Here are the key components of a feasibility study:

1. Operational Feasibility

- The project addresses the need for assistive technology, providing a user-friendly system for individuals with mobility impairments.
- Support materials will be available to help users understand and effectively use the system.

2. Technical Feasibility

- The integration of EEG and EOG signal acquisition via two BioAmp EXG Pills, along with real-time processing capabilities, ensures the system's technical feasibility.
- Open-source libraries and frameworks facilitate efficient development and troubleshooting.

3. Economic Feasibility

- The cost of hardware components (Arduino, BioAmp EXG Pill, electrodes) remains within budget, and open-source software tools minimize costs.
- The system's potential for long-term applications in assistive technology adds significant value.

3.2 Design

Software design is the process of creating a detailed blueprint for implementing software based on analyzed requirements. It defines how various components, modules, and interactions will work together to achieve the desired functionality. The purpose of software design is to provide a clear structure for development, ensure scalability and maintainability, reduce risks by identifying potential issues early, and facilitate smoother implementation, testing, and debugging. Additionally, it helps enhance user experience through intuitive interfaces and reliable system performance.

3.2.1 Flowchart

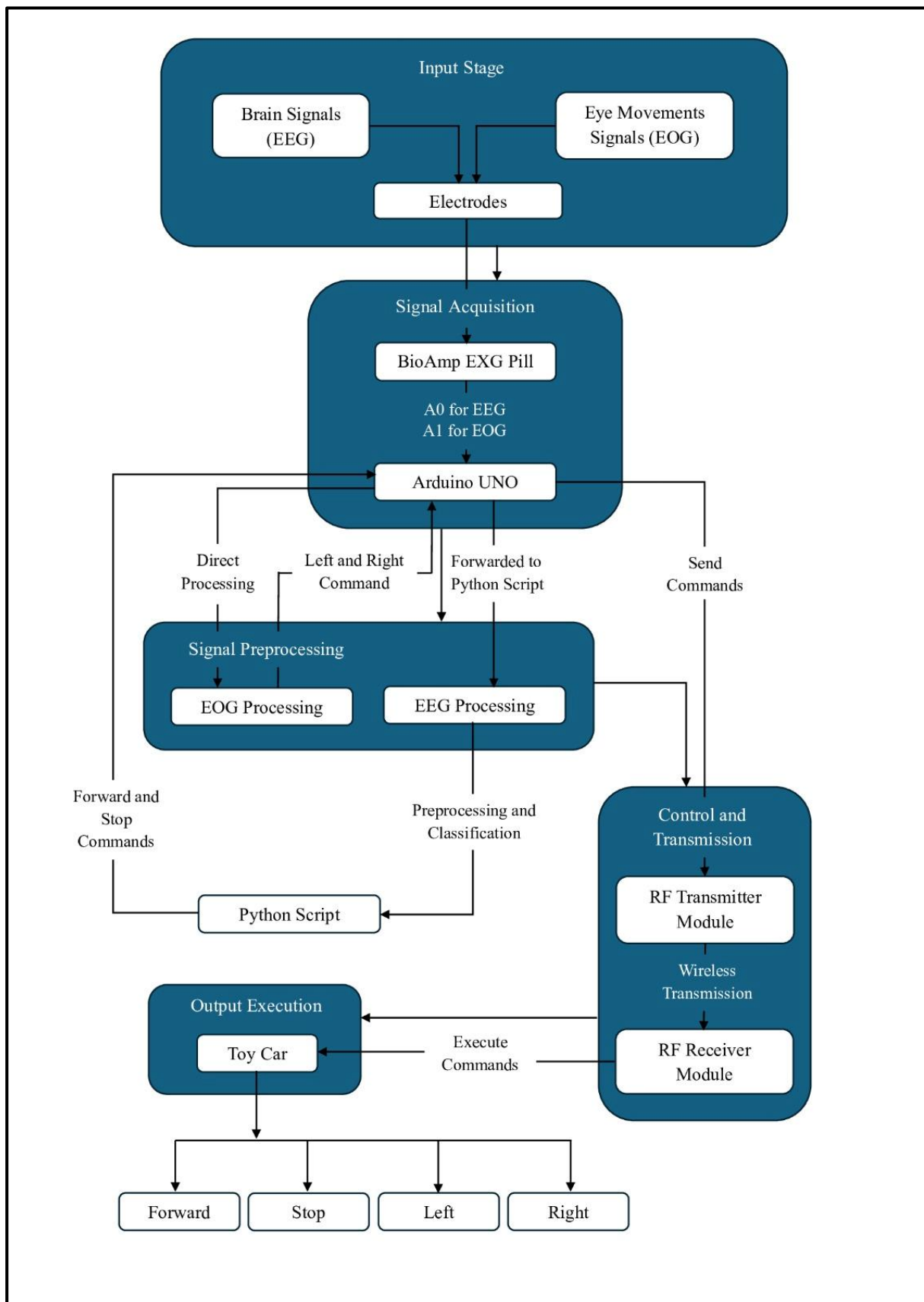


Figure 3.2.1 Model Implementation Flowchart

3.2.2 UML Diagrams

1. Data Flow Diagram (Context Level)

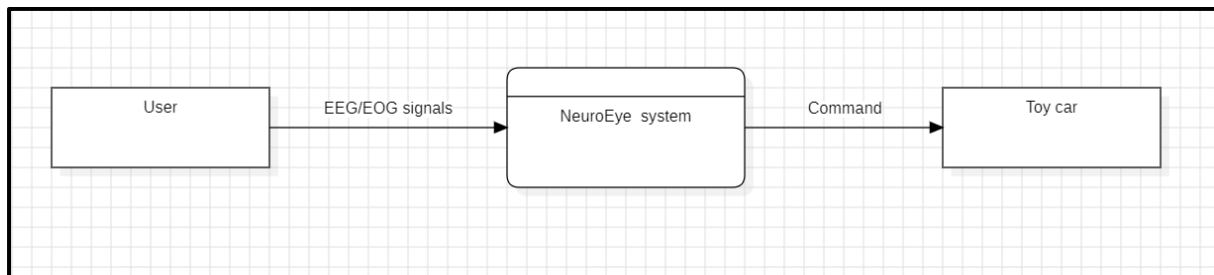


Figure 3.2.2 1) Data Flow Diagram (Context Level)

2. Data Flow Diagram (Level 1)

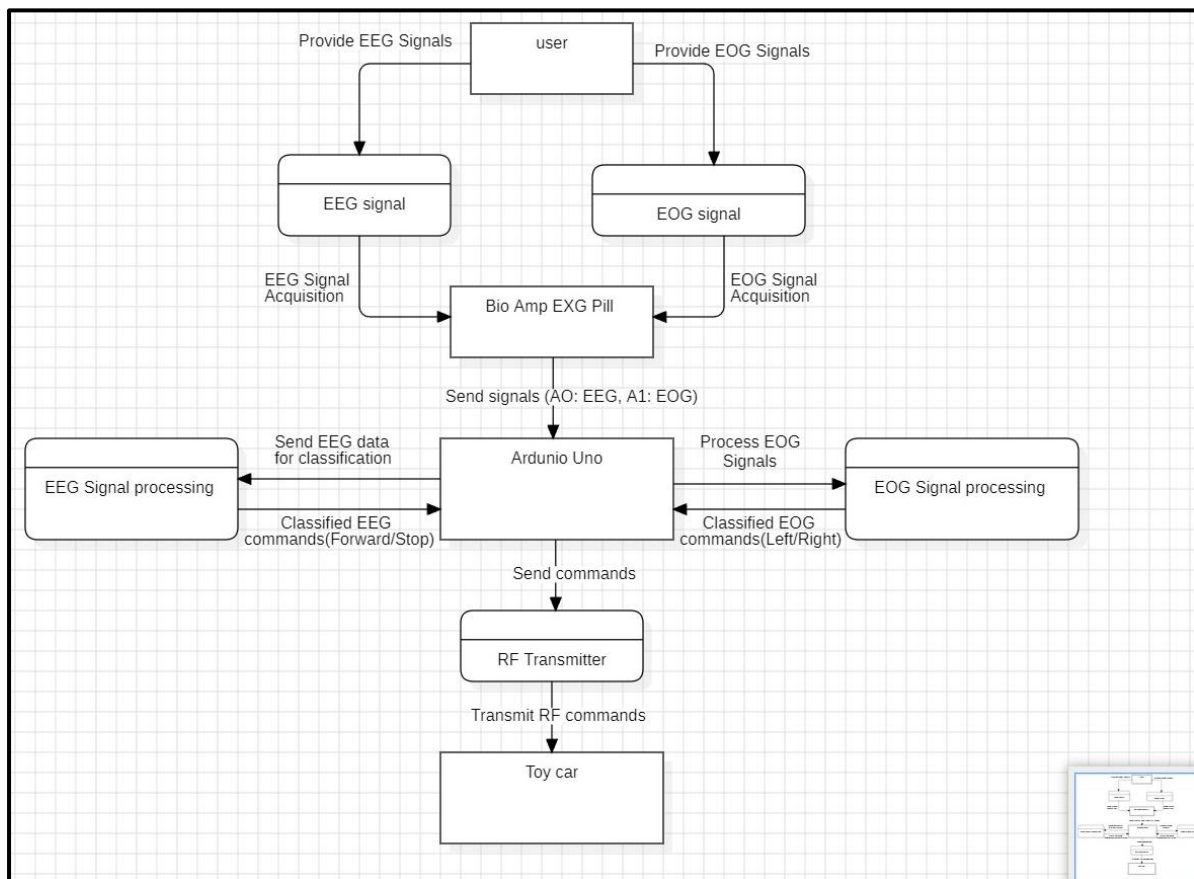


Figure 3.2.2 2) Data Flow Diagram (Level 1)

3. Data Flow Diagram (Level 2)

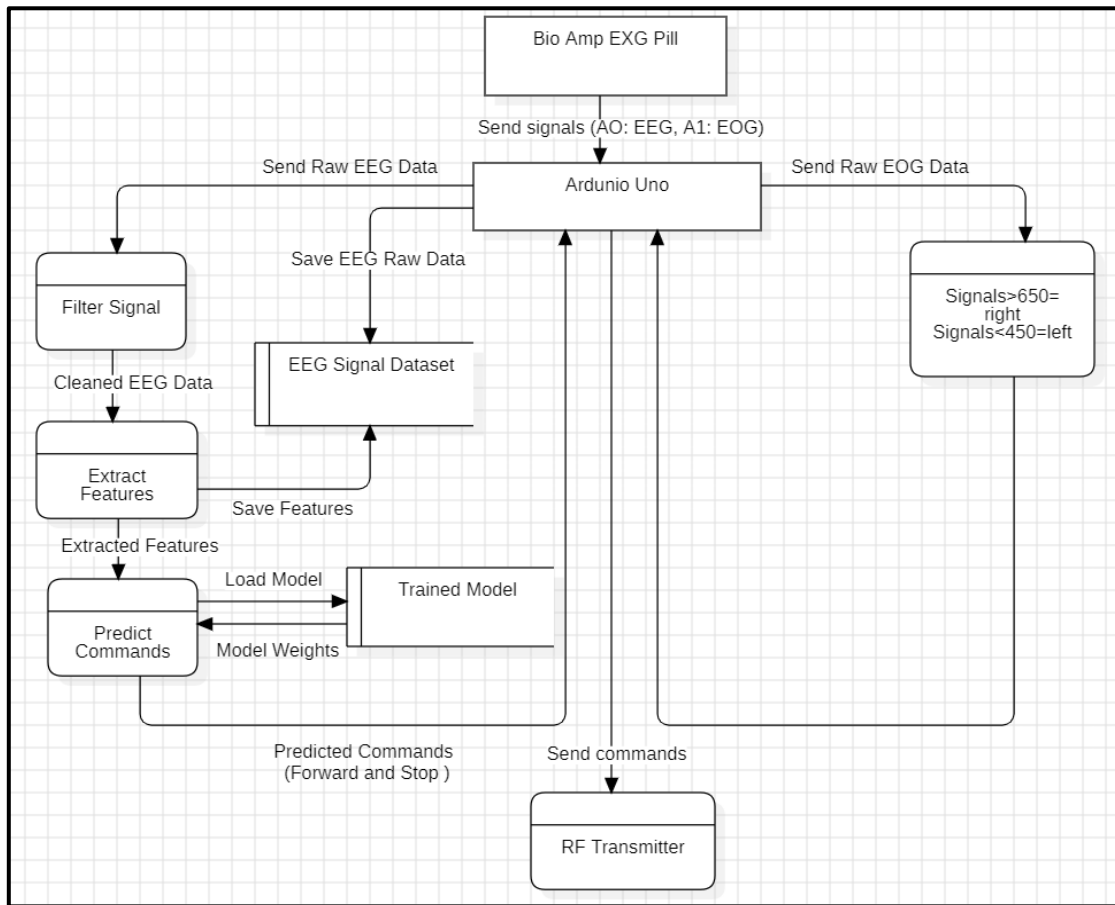


Figure 3.2.2 3) Data Flow Diagram (Level 2)

4. Sequence diagram

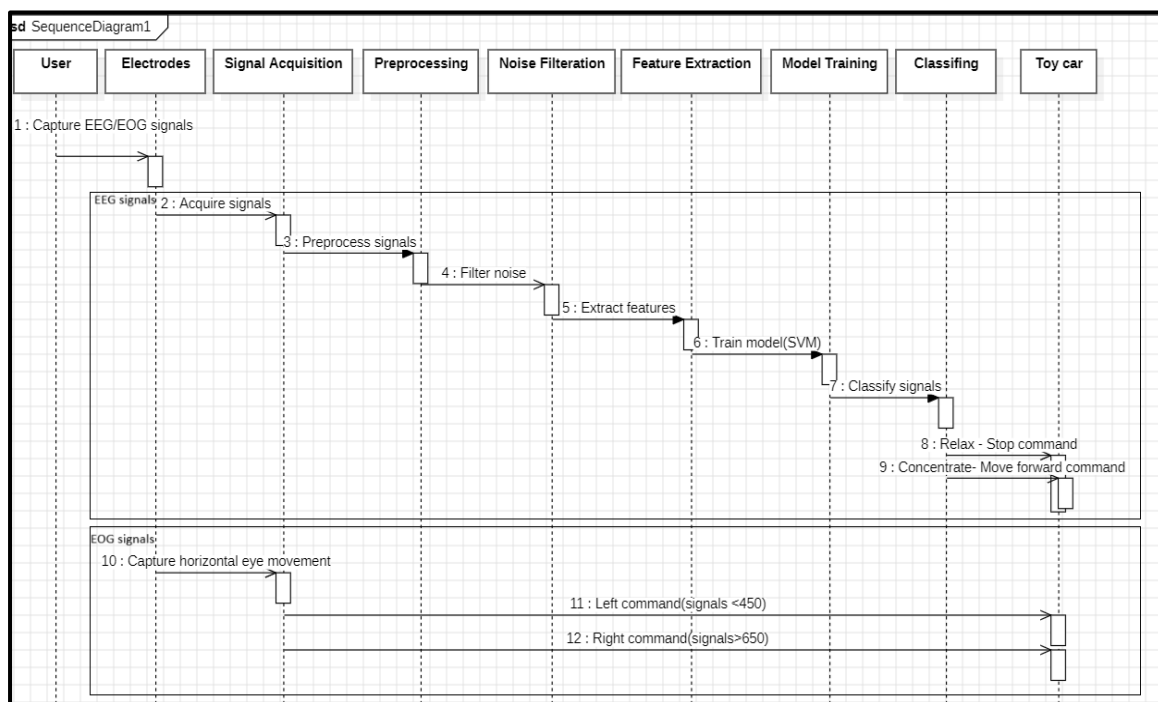


Figure 3.2.2 4) Sequence diag

CHAPTER 4

SYSTEM IMPLEMENTATION & TESTING

4.1 For EEG Signals:

1. Signal Acquisition

The process of acquiring EEG signals involves multiple steps to ensure high-quality data for analysis:

- **Electrode Placement:**
 - Forehead Electrodes: Positioned on the forehead to detect brain activity associated with specific mental states, such as relaxation and concentration.
 - Earlobe Electrodes: Used as reference electrodes to minimize ambient electrical noise, improving the overall signal quality.
- **Signal Amplification:**
 - The BioAmp EXG Pill is used to amplify the weak electrical signals from the brain, making them suitable for further processing and analysis.
- **Sampling Rate:**
 - The sampling rate is set to 512 Hz, which is a standard for EEG signal processing.
 - This rate captures key brainwave patterns, including delta (0.5–3 Hz), theta (4–7 Hz), alpha (8–13 Hz), and beta (14–30 Hz) frequencies, ensuring the preservation of meaningful neural activity.
- **Real-Time Signal Reading:**
 - Signals are acquired in real-time using serial communication with the serial.Serial interface.
- **Buffering:**
 - Incoming EEG values are stored in a circular buffer implemented with Python's deque.
 - The buffer size is set to 512 samples, corresponding to a 1-second window of EEG data.
 - This approach ensures that only the most recent data is analyzed, maintaining a sliding window for real-time signal processing.

2. Data preprocessing

2.1 Filtration

Filtration is a crucial step in EEG signal processing, ensuring the data is clean, interpretable, and relevant for analysis.

- **Notch Filtering:** This is critical to eliminate interference commonly found in EEG recordings.
- **Bandpass Filtering:** Retains only frequencies between 0.5 Hz and 30 Hz, which include key EEG bands (delta, theta, alpha, beta). Filters out irrelevant low-frequency drift and high-frequency noise.

```
def setup_filters(sampling_rate):
    b_notch, a_notch = signal.iirnotch(50.0 / (0.5 * sampling_rate), 30.0)
    b_bandpass, a_bandpass = signal.butter(4, [0.5 / (0.5 * sampling_rate), 30.0 / (0.5 *
sampling_rate)], 'band')
    return b_notch, a_notch, b_bandpass, a_bandpass
```

This function designs the filters required for processing EEG data. Filtering helps to remove noise and retain relevant signal components.

- Inputs: sampling_rate (rate at which EEG data is sampled, e.g., 512 Hz).
- Filters:
 - 1) **Notch filter (iirnotch):**
 - Removes 50 Hz powerline interference (common in electrical systems).
 - Parameters:
 - 50.0 / (0.5 * sampling_rate): Normalized frequency of the notch filter.
 - 30.0: Quality factor, determining filter sharpness.
 - 2) **Bandpass filter (butter):**
 - Keeps frequencies between 0.5 Hz and 30 Hz, which correspond to brainwave signals (delta, theta, alpha, beta).
 - Parameters:
 - 4: Filter order (determines sharpness).
 - [0.5 / (0.5 * sampling_rate), 30.0 / (0.5 * sampling_rate)]: Normalized frequency range.

2.2 Processing EEG signals

```
def process_eeg_data(data, b_notch, a_notch, b_bandpass, a_bandpass):
    data = signal.filtfilt(b_notch, a_notch, data)
    data = signal.filtfilt(b_bandpass, a_bandpass, data)
    return data
```

This function applies the filters designed in the previous step to raw EEG data.

- Inputs:
 - data: The raw EEG signal array (512 samples).
 - Filter coefficients: b_notch, a_notch for the notch filter and b_bandpass, a_bandpass for the bandpass filter.
- Processing Steps:
 - First step: Apply the notch filter using filtfilt, which removes 50 Hz noise without introducing phase distortion.
 - Second step: Apply the bandpass filter to isolate relevant brainwave frequencies.
- Output: Returns the filtered EEG signal for further analysis.

3. Feature Extraction

Feature extraction transforms the processed signal into meaningful numerical representations for classification.

Frequency-Domain Features (Power Spectral Density - PSD)

Power Spectral Density (PSD) represents the distribution of power (signal strength) of an EEG signal across different frequencies. It is a frequency-domain feature that provides insights into how signal energy is distributed over the frequency spectrum.

```
def calculate_psd_features(segment, sampling_rate):  
    f, psd_values = signal.welch(segment, fs=sampling_rate, nperseg=len(segment))  
    bands = {'alpha': (8, 13), 'beta': (14, 30), 'theta': (4, 7), 'delta': (0.5, 3)}
```

This function computes features from EEG signals using **Welch's method** for PSD. Key steps:

1. **PSD Calculation:** Computes frequency bins (f) and power values (psd_values) for EEG segments.
2. **Frequency Bands:** Focuses on Alpha (8–13 Hz), Beta (14–30 Hz), Theta (4–7 Hz), and Delta (0.5–3 Hz).
3. **Feature Extraction:** Calculates total power for each band (E_alpha, E_beta, etc.) and computes the Alpha/Beta ratio to indicate mental states like relaxation or focus.

Additional Features:

```
def calculate_additional_features(segment, sampling_rate):  
    f, psd = signal.welch(segment, fs=sampling_rate, nperseg=len(segment))  
    peak_frequency = f[np.argmax(psd)]  
    spectral_centroid = np.sum(f * psd) / np.sum(psd)  
    spectral_slope = np.polyfit(log_f, log_psd, 1)[0]
```

- **Peak Frequency:**
 - The frequency with the highest power, indicating dominant brain activity.
- **Spectral Centroid:**
 - The "center of mass" of the power spectrum, providing a summary of frequency distribution.
- **Spectral Slope:**
 - Indicates the rate at which power decreases with increasing frequency. It is calculated by fitting a line to the log-log PSD data.

4. Data Standardization

```
scaler = StandardScaler()  
X_train_scaled = scaler.fit_transform(X_train)
```

- Standardization ensures that the model receives input with a consistent range and distribution, improving its accuracy.
- The StandardScaler is used to standardize the features (X) to have a mean of 0 and a standard deviation of 1.

5. Machine Learning Model Used:

For model training, we use the SVC (Support Vector Classification) model from sklearn.svm.

```
clf = SVC(kernel='linear', random_state=42)  
clf.fit(X_train_scaled, y_train)
```


Why SVM?

- SVMs are effective for high-dimensional data like EEG features.
- They perform well with limited training data, a typical scenario for EEG systems.

4.2 For EOG Signals:

1. Signal Acquisition:

- Electrodes are positioned near the outer corners of the eyes to capture horizontal eye movements. This placement ensures effective detection of gaze direction.
- EOG signals are recorded in real-time, synchronized with EEG recordings, to align both data types during analysis.

2. Signal Processing:

```
#define THRESHOLD_LEFT 450
#define THRESHOLD_RIGHT 650
#define DEBOUNCE_DELAY 2000
if (currentTime - lastMovementTime >= DEBOUNCE_DELAY) {
    if (eog_signal < THRESHOLD_LEFT && !movementDetected) {
        // Left movement
        left();
        delay(600);
        stops();
        movementDetected = true;
        lastMovementTime = currentTime;
    } else if (eog_signal > THRESHOLD_RIGHT && !movementDetected) {
        // Right movement
        right();
        delay(600);
        stops();
        movementDetected = true;
        lastMovementTime = currentTime;
    }
}
```

Thresholds for EOG (Eye Movements):

- **THRESHOLD_LEFT (450):** If the EOG signal is below this value, it indicates a left eye movement.
- **THRESHOLD_RIGHT (650):** If the EOG signal is above this value, it indicates a right eye movement.
- **DEBOUNCE_DELAY (2000 ms):** Prevents the car from reacting too quickly to small fluctuations in the eye movement signal.

4.3 Hardware Setup

4.3.1. Circuit Diagram

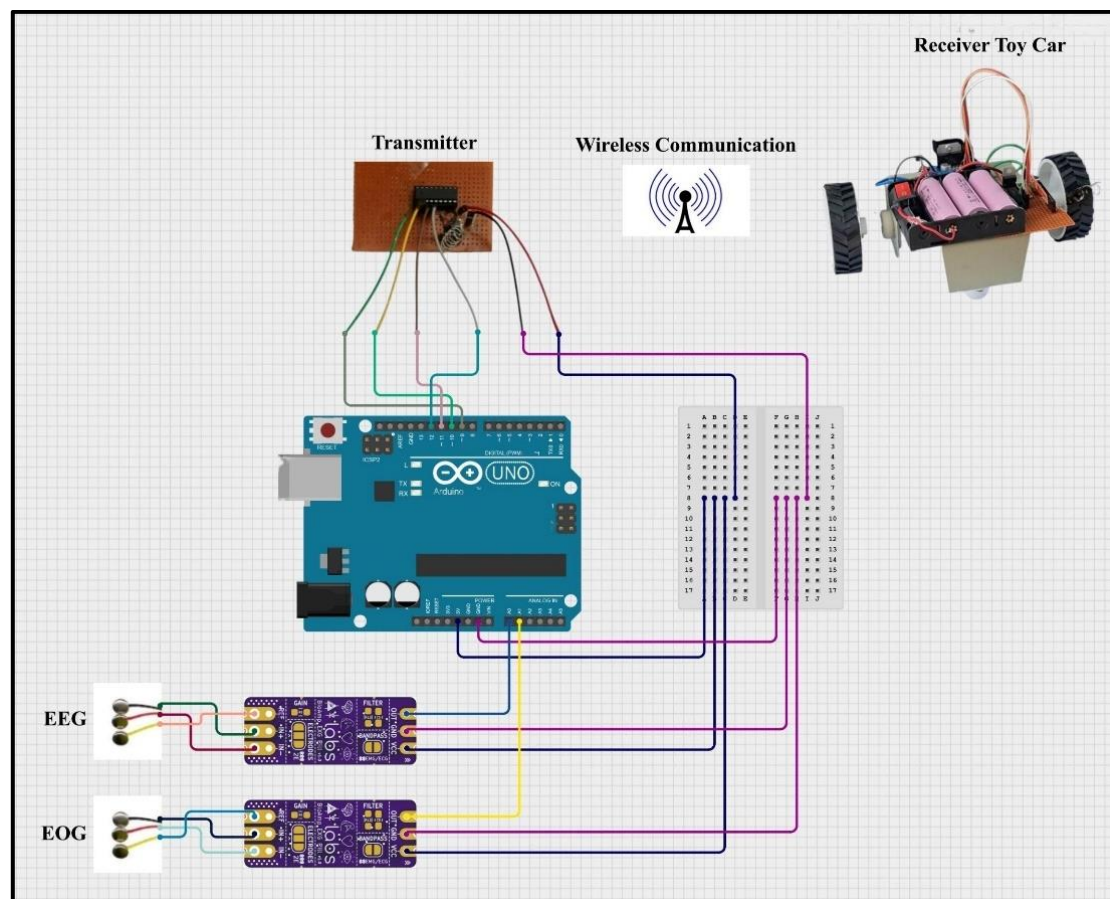


Figure 4.3.1. Circuit Diagram

Connections:

1. EEG and EOG Modules

- **Signal Outputs:**
 - The EEG and EOG signal lines are connected to the analog input pins on the Arduino (e.g., A0 for EEG, A1 for EOG).
- **Power Supply:**
 - **VCC (Power):** Connected to the 5V pin on the Arduino.
 - **GND (Ground):** Connected to the GND pin on the Arduino.

2. Arduino UNO

- **Analog Inputs:**
 - Receives signals from the EEG module (e.g., A0) and the EOG module (e.g., A1).
- **Digital Outputs:**
 - Sends processed control signals to the transmitter module via digital pins (e.g., D10, D11).
- **Power Supply:**
 - Powered either through USB or via an external power source.

3. Transmitter Module

- **Input Lines:**
 - Connected to the digital output pins on the Arduino (e.g., D10, D11).
- **Power Supply:**
 - **VCC (Power):** Connected to the 5V pin on the Arduino.
 - **GND (Ground):** Connected to the GND pin on the Arduino.

4. Breadboard

- Serves as a connection hub, distributing power and signals between the Arduino and the various modules.

5. Toy Car (Receiver)

- **Receiver Input:**
 - Paired with the transmitter module, it receives the control signals wirelessly.
- **Motor Driver:**
 - Connects to the motors of the toy car, enabling movement based on the received control signals.

4.3.2. Code Implementation (Arduino)

loop () Function:

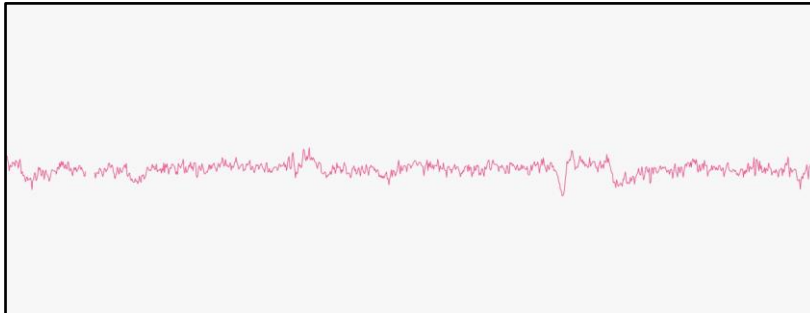
- **EEG Signal Processing (Forward/Stop Control):**
 - A timer is used to control the sampling rate of the EEG signal. It reads the EEG signal from EEG_PIN every time the timer reaches 0 (based on SAMPLE_RATE).
 - The EEG signal is sent to Python via serial communication.
 - If Python sends a prediction:
 - If the prediction is "1", it calls forward() to move the car forward.
 - If the prediction is "0", it calls stops() to stop the car.
- **EOG Signal Processing (Left/Right Movement Control):**
 - The EOG signal is read from EOG_PIN. The eye movement is detected if the signal exceeds predefined thresholds:
 - If the signal is below THRESHOLD_LEFT, the car turns left by calling left().
 - If the signal is above THRESHOLD_RIGHT, the car turns right by calling right().

4.3.3. Integrating Trained Model and Hardware

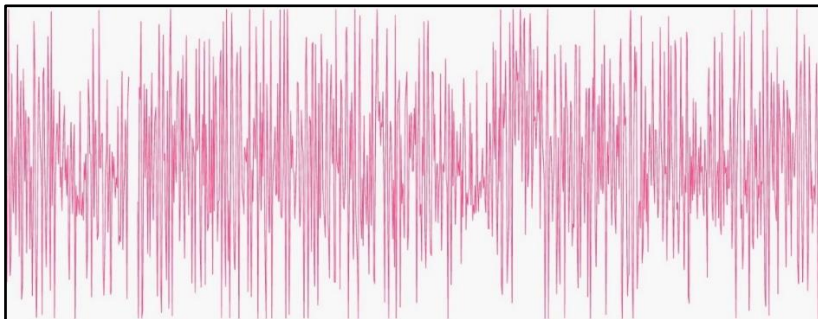
1. **Training the Model:** A machine learning model (e.g., SVM, as discussed) is trained offline in Python using EEG data. The trained model is stored in a .pkl file (model.pkl).
2. **Data Exchange:** The Arduino sends EEG data to the Python script, where the model predicts the user's intent (e.g., forward or stop). The Python script sends this prediction back to Arduino via serial communication.
3. **Car Control:** The Arduino receives the prediction and controls the car's movement accordingly (forward or stop) by activating the appropriate LEDs or motors.

4.4 Results

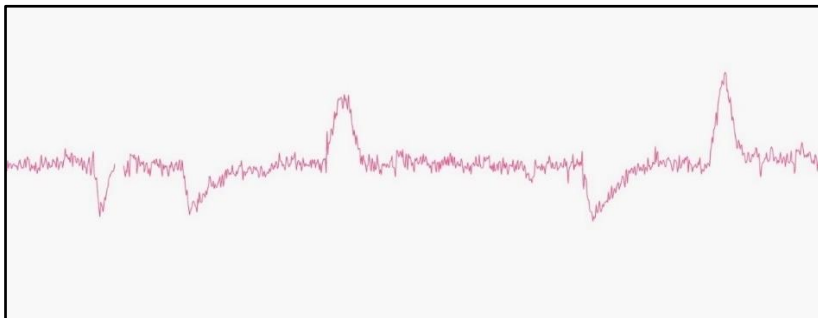
4.4.1. EEG Signal for “Stop” Command (Relaxed State)



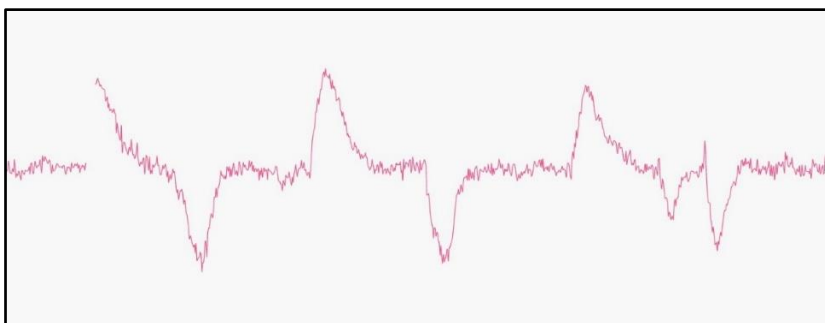
4.4.2. EEG Signals for “Forward” Command (Concentrate state)



4.4.3 EOG Signals for “Left” Command (Left Horizontal Eye Movement)



4.4.4 EOG Signals for “Right” Command (Right Horizontal Eye Movement)



4.5 Output Images

4.5.1 Stop command:



Figure 4.5.1 i) User in relaxed mode

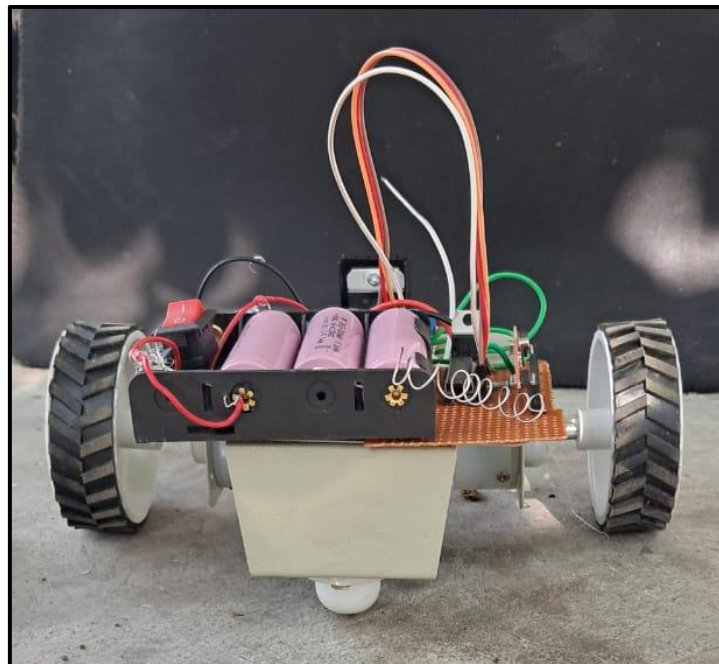


Figure 4.5.1 ii) Toy car in stop position

4.5.2 Forward command:



Figure 4.5.2 i) User in concentrated mode

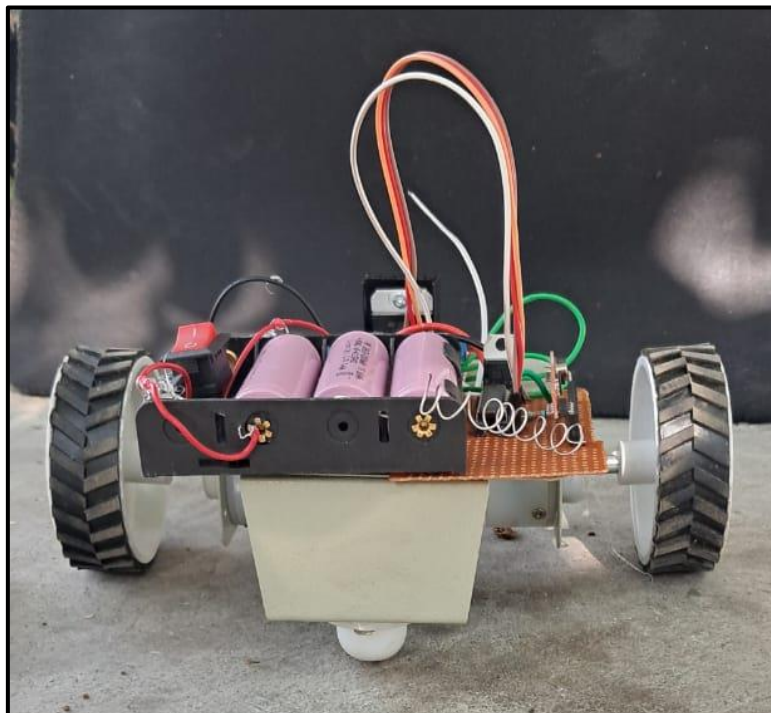


Figure 4.5.2 ii) Toy car moving forward

4.5.3 Left command:



Figure 4.5.3 i) Left Horizontal Eye Movement

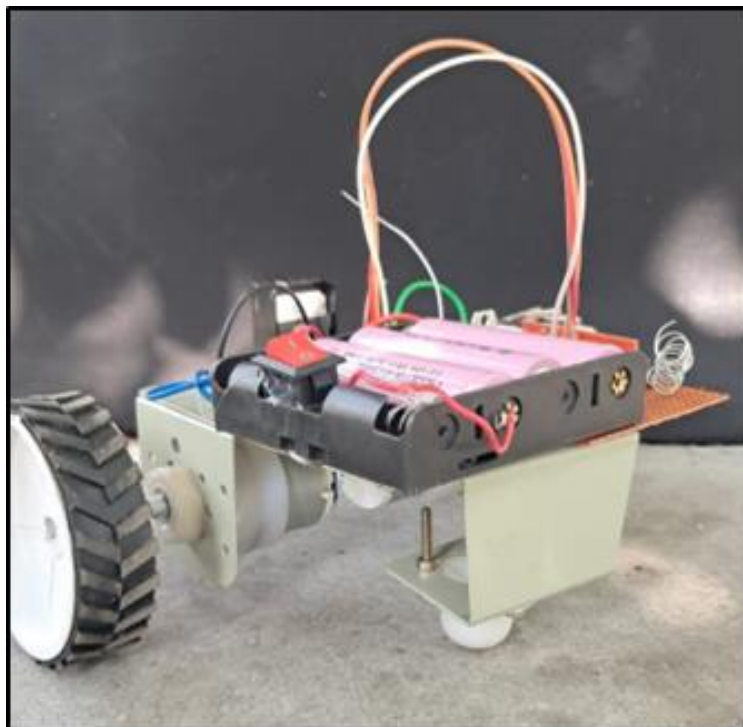


Figure 4.5.3 ii) Toy car moving in left direction

4.5.4 Right command:



Figure 4.5.4 i) Right Horizontal Eye Movement

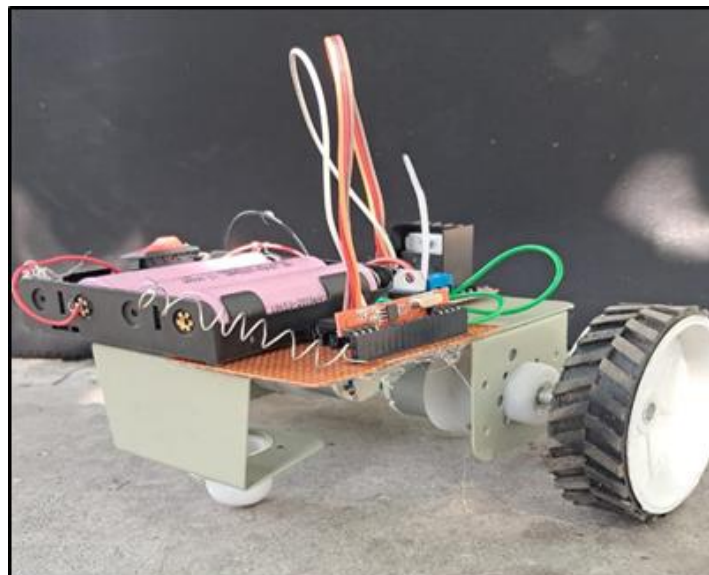


Figure 4.5.4 ii) Toy car moving in right direction

4.6 Testing

1. Unit Testing

- **Objective:** Test individual components like EEG and EOG modules separately.
- **Performed Tests:**
 - **EEG Module:** Ensured proper acquisition of brainwave signals, focusing on command signals for "forward," "stop," etc.
 - **EOG Module:** Verified accurate detection of eye movements (e.g., left, right, up, down) using electrodes near the eyes.
 - **Signal Processing Module:** Tested individual preprocessing steps, such as filtering EEG/EOG noise and feature extraction, to ensure clean input for classification.
 - **Command Mapping Module:** Checked the translation of EEG and EOG signals into actionable commands for controlling the toy car.

2. Integration Testing

- **Objective:** Ensure seamless interaction between EEG and EOG modules, as well as signal processing and control systems.
- **Performed Tests:**
 - **EEG and EOG Synchronization:** Verified that signals from both sources are correctly aligned and interpreted without significant delays.
 - **Combined Signal Processing:** Ensured that preprocessing algorithms (e.g., for noise reduction) work simultaneously for EEG and EOG inputs.
 - **Command Execution:** Tested the system end-to-end by combining EEG commands (e.g., forward, stop) with EOG navigation (e.g., left, right) to drive the toy car.

3. Alpha Testing

- **Objective:** Internal team testing to identify system-level bugs.
- **Performed Tests:**
 - Evaluated the system under controlled environments to ensure all commands (EEG and EOG) were correctly interpreted and executed.
 - Simulated multiple scenarios, such as prolonged use and varying signal intensities, to detect reliability issues.

- Repeatedly tested synchronization between EEG and EOG signals to ensure no misinterpretation during simultaneous use.

4. Beta Testing

- **Objective:** External testing with selected users to evaluate real-world usability.
- **Performed Tests:**
 - Users tested the toy car in real-world environments with varying lighting and noise levels.
 - Collected feedback on the system's ease of use, response time, and command accuracy.
 - Tested for user adaptability and fatigue to identify potential improvements in the interface or processing.

5. White-Box Testing

- **Objective:** Evaluate the internal structure and logic of algorithms and code.
- **Performed Tests:**
 - Reviewed and debugged signal processing algorithms to ensure EEG and EOG signals were processed without interference.
 - Verified classification logic for both EEG and EOG signals, ensuring that specific brainwave patterns and eye movements map to the correct commands.
 - Assessed the system's error-handling mechanisms for cases where signals were incomplete or ambiguous.

6. Black-Box Testing

- **Objective:** Test the system functionality from the user's perspective.
- **Performed Tests:**
 - Provided inputs through EEG and EOG and verified if the expected outputs (e.g., toy car movement in a specific direction) matched user intent.
 - Assessed how the system handles invalid or ambiguous signals, such as unintentional eye blinks or weak EEG patterns.
 - Simulated different use cases, like rapid changes in command inputs, to ensure stability and accuracy.

7. Technical Feasibility

Objective: Verify that the Arduino, EEG, EOG, BioAmp Pill, and RF module work together effectively.

Tests Conducted:

- Signal Acquisition: Verified Arduino's ability to read EEG/EOG signals from the BioAmp Pill.
- Preprocessing: Tested Arduino's ability to preprocess and filter the signals.
- Signal Transfer: Assessed Arduino's ability to send processed signals to a computer via serial communication.
- Real-Time Control: Evaluated the real-time response of controlling the toy car using EEG/EOG signals.

Findings:

- Arduino can acquire and preprocess basic EEG/EOG signals but lacks computational power for advanced processing.
- Serial communication works well for sending signals to a computer.
- Real-time control is feasible.

8. Economic Feasibility

Objective: Ensure the system remains cost-effective.

Tests Conducted:

- Compared costs of Arduino-based setups with alternatives.
- Assessed cost reduction via open-source libraries.

Findings:

- The Arduino is cost-effective, minimizing hardware costs and eliminating the need for high-end processors.

CHAPTER 5

FUTURE SCOPE AND CONCLUSION

5.1. Applications

1. **Assistive Technologies:** Can be adapted for wheelchairs and mobility aids to enhance the independence of individuals with disabilities.
2. **Gaming Interfaces:** Offers a unique way to control video games using thoughts and eye movements, providing an immersive experience.
3. **Smart Home Devices:** Integration with smart home systems allows users to control lights, appliances, and other devices using brain signals.
4. **Robotics:** Applicable in controlling robotic arms or drones, enabling precise manipulation of objects through brain and eye commands.
5. **Healthcare Monitoring:** Can be used in rehabilitation settings to monitor patients' mental states during therapy sessions.
6. **Research and Development:** Useful for BCI research to study brain activity and improve signal processing techniques.
7. **Education Tools:** Potential for developing educational tools that adapt to students' concentration levels and learning paces.
8. **Virtual Reality:** Integration with VR systems for a more engaging and responsive user experience in virtual environments.
9. **Elderly Assistance:** Helps elderly individuals control devices without physical effort, improving their quality of life.
10. **Mental Health Applications:** Can be used to monitor and analyze emotional states, aiding in mental health assessments and treatments.

5.2. Advantages:

1. **Non-Invasive:** Utilizes non-invasive methods for signal acquisition, ensuring safety and comfort for users.
2. **Enhanced Independence:** Empowers users with mobility impairments to control devices without physical effort, promoting autonomy.
3. **Intuitive Control:** Provides a natural way to interact with technology through thoughts and eye movements, enhancing user experience.
4. **Real-Time Processing:** Ensures commands are executed with minimal latency, allowing for responsive control of devices.
5. **Customizable Interface:** Users can personalize control settings according to their individual needs and preferences.

6. **Broad Application Potential:** Adaptable for various applications, from entertainment to healthcare, making it versatile.
7. **Cost-Effective Solution:** Compared to invasive BCI systems, this approach is more affordable and accessible.
8. **User-Friendly:** Designed to be easy to use, minimizing the learning curve for new users.
9. **Promotes Engagement:** Encourages active participation in various tasks, improving cognitive function and engagement.
10. **Scalability:** Potential to scale the technology for larger applications, such as controlling multiple devices simultaneously.

5.3. Future Scope

1. **Advanced Control Systems:** Explore integration with additional sensors (e.g., heart rate monitors) for improved navigation and control accuracy.
2. **Machine Learning Enhancements:** Implement advanced machine learning algorithms to refine signal classification and processing for greater accuracy.
3. **Wireless Technology:** Investigate wireless communication methods for signal transmission to enhance user mobility and reduce setup complexity.
4. **User Feedback Systems:** Develop systems to provide real-time feedback to users based on their mental states for improved interaction.
5. **Scalability:** Potential to scale the technology for broader applications, such as controlling multiple devices or systems simultaneously.
6. **Community Engagement:** Involve users in the design process to better understand their needs and improve the system's usability.
7. **Multi-User Systems:** Explore the possibility of accommodating multiple users, allowing for shared control of devices.
8. **Integration with AI:** Utilize artificial intelligence to predict user commands based on historical data, enhancing responsiveness.
9. **Accessibility Features:** Develop additional features to support users with varying levels of disabilities, ensuring inclusivity.
10. **Longitudinal Studies:** Conduct long-term studies to assess the effectiveness and impact of BCI technology on users' quality of life.

5.4. Conclusion

The “Neuro-Eye Controlled Toy Car” project showcases the effective integration of EEG and EOG signals to facilitate intuitive control of a toy car, illustrating the practical applications of brain-computer interface (BCI) technology. By allowing users to command the car's movements through mental states and eye movements, this initiative significantly enhances the independence of individuals with mobility impairments. The project not only validates the feasibility of non-invasive BCI systems but also emphasizes the importance of user-friendly interfaces that can adapt to various needs, ensuring a seamless interaction between the user and technology.

Furthermore, the findings from this research pave the way for future advancements in assistive technologies and BCI applications. By demonstrating the potential for real-time command execution through mental concentration and eye movements, the project opens up new avenues for the development of sophisticated devices, including mobility aids and smart home systems. As the field of BCI continues to evolve, this study contributes valuable insights that can inspire further innovation, ultimately improving the quality of life for individuals with disabilities and expanding the capabilities of BCI technology in everyday life.

ANNEXURE

ANNEXURE – A: TCS iON Career Edge – Young Professionals



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Mehul Mehta
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Mehul Mehta

Mehul Mehta
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Start Date: 29 Sep 2024

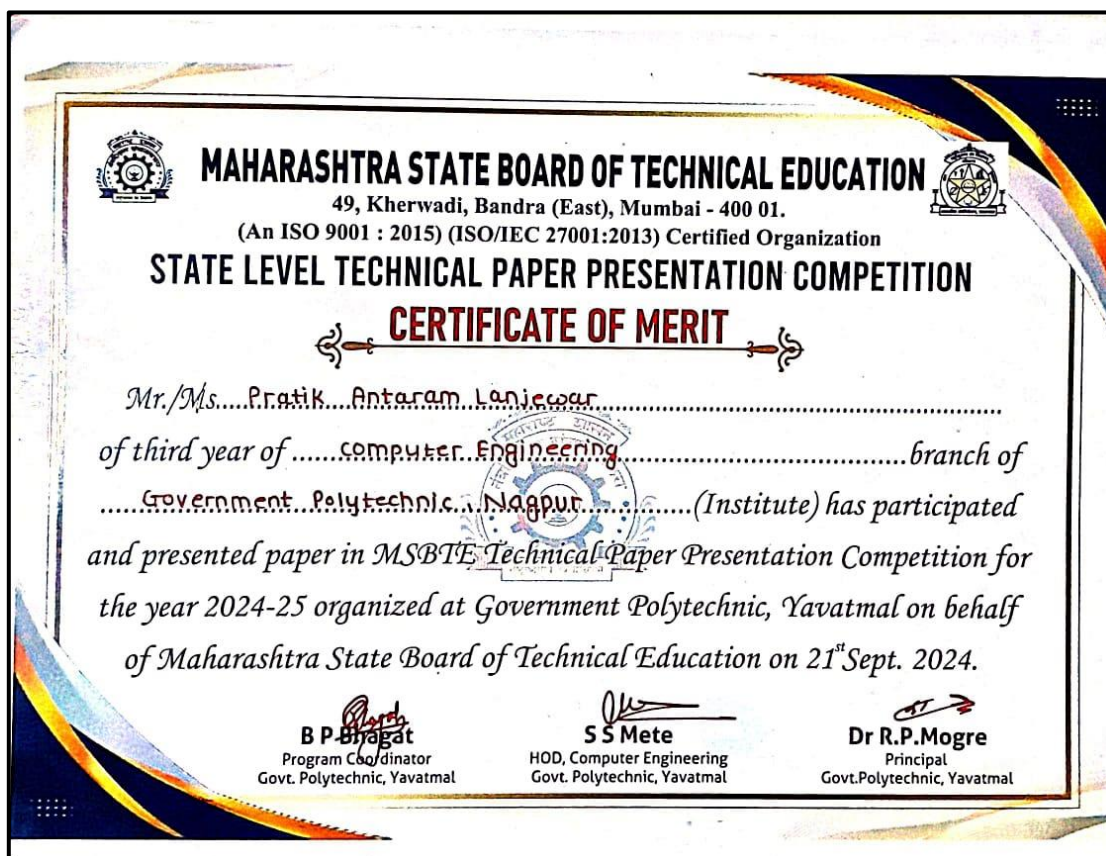
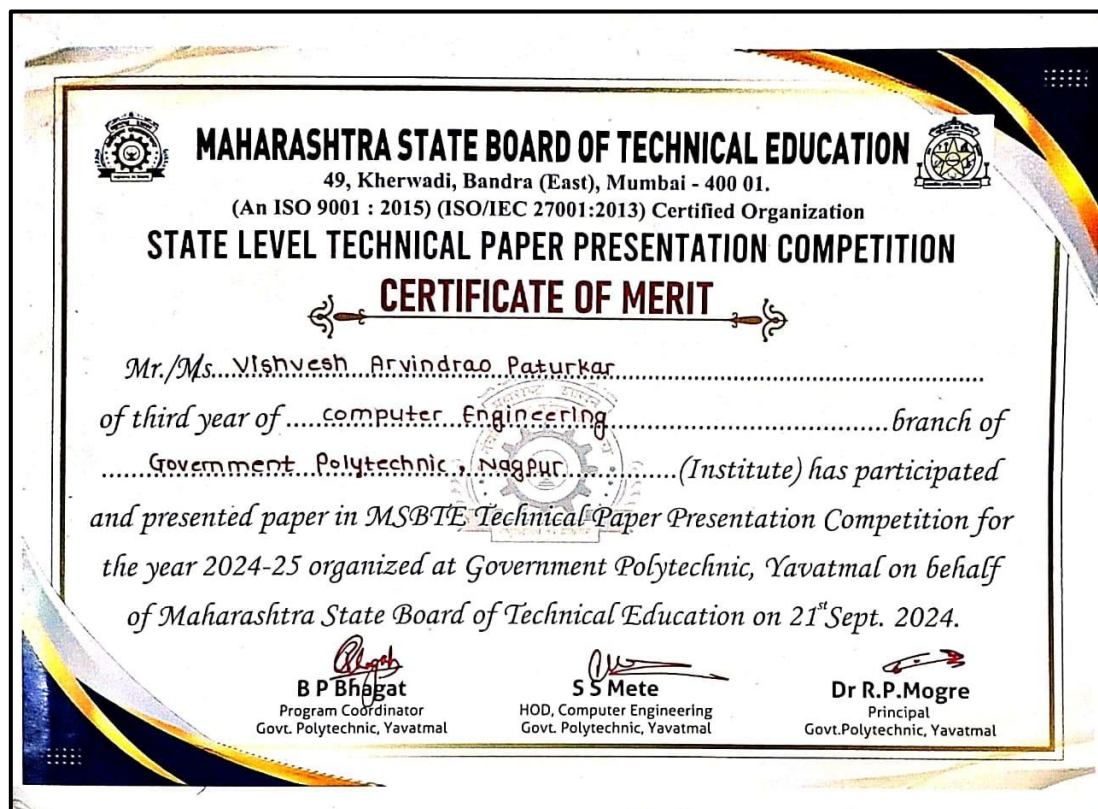
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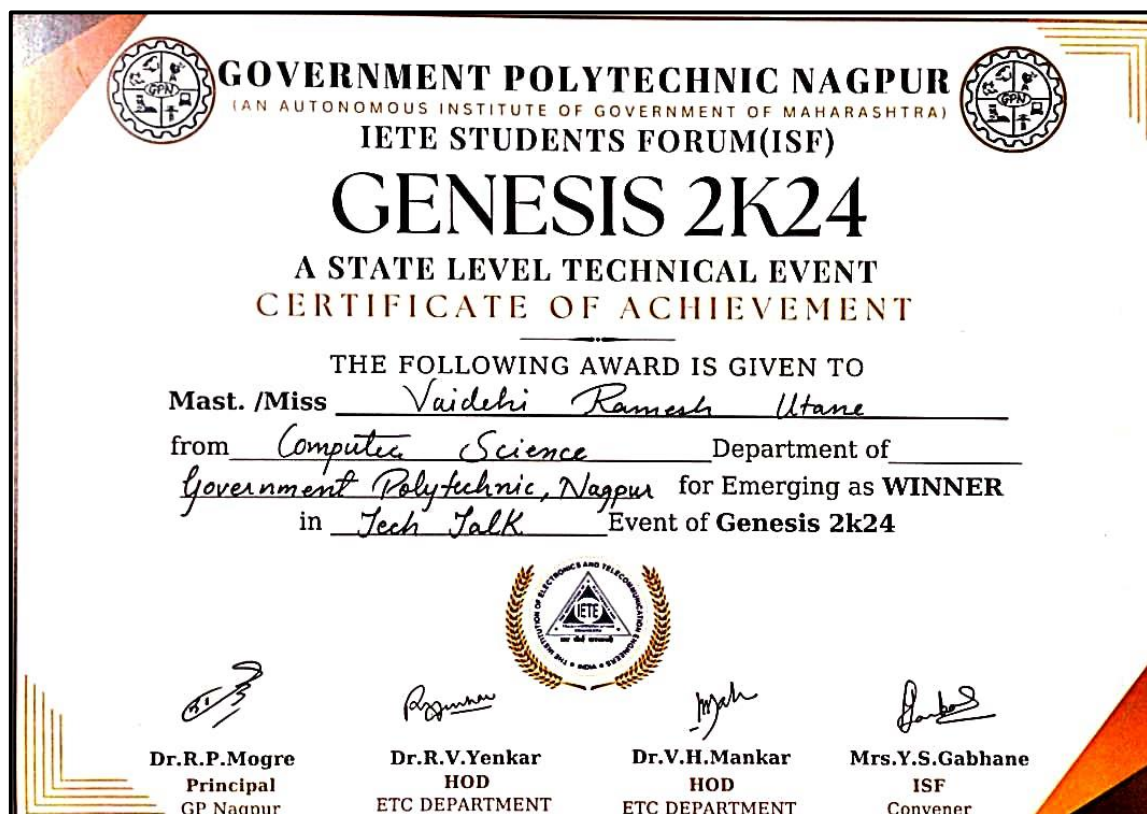
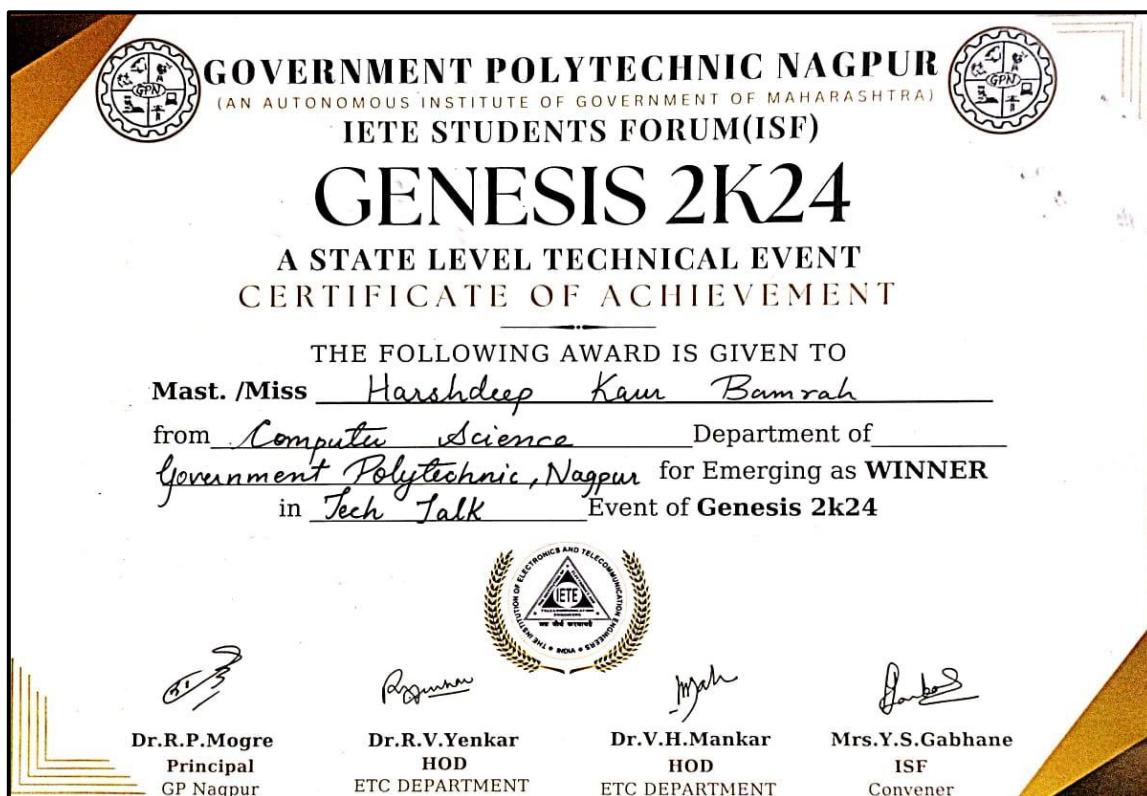


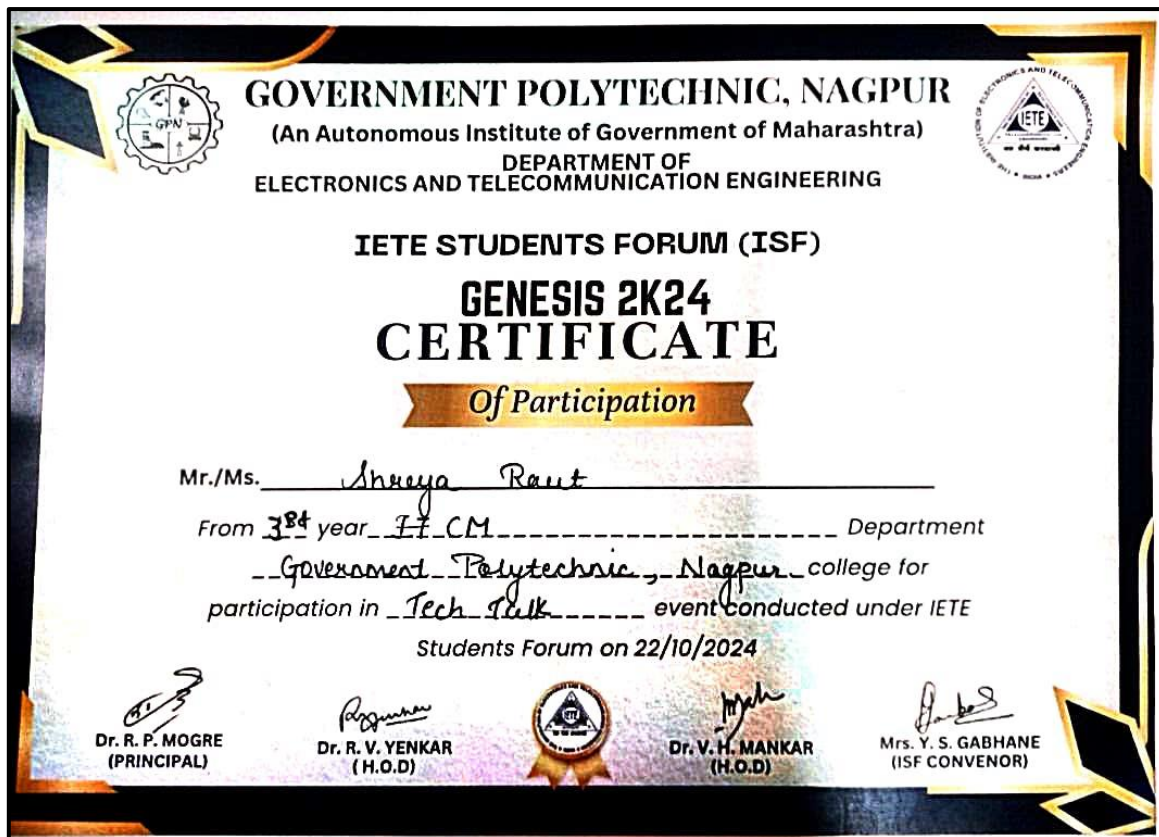
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ANNEXURE – B : State Level Technical Paper Presentation Competition







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- [8] Visual Studio Code: <https://code.visualstudio.com/download>
- [9] StarUML: <https://staruml.io/>
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